

Tone 3 Sandhi and Its Interaction with Stress in Mandarin

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Abstract

This paper documents Tone 3 sandhi in Mandarin Chinese, a Sino-Tibetan language spoken widely across China and the Chinese diaspora. The data come from elicitation sessions with a native Mandarin speaker and largely confirm the findings of Lin (2007), while offering new evidence on how stress affects the application of Tone 3 sandhi in phrases.

Lin describes Tone 3 sandhi in Mandarin as a tonal alternation in which a Tone 3 (214) alternates into a Tone 2 (35) when followed by another Tone 3, as in /laʊ35 ʂu/214 → [laʊ35 ʂu214] ‘mouse’. Tone 3 reduction, another related tonal alternation he describes, is the case where a T3 is reduced or shortened (214 → 21) when they are followed by another tone. The basic domain for Tone 3 sandhi is the disyllabic foot which starts with the smallest morpho-syntactic domain and could extend both forward or backward depending on the phrase structure. For example, phrases like /mi214 laʊ214 ʂu214 xɑʊ214/ alternates to [mi214 laʊ35 ʂu35 xɑʊ214] ‘Mickey Mouse is good’ because the ‘mouse’ is the smallest morpho-syntactic domain in this phrase and Tone 3 sandhi starts with that domain.

The speaker for this paper applies Tone 3 sandhi consistently and naturally, aligning with Lin’s findings. When asked to stress different words in the same phrase (e.g., “the small FEMALE horse is good” vs. “the small female HORSE is good”), the speaker attempts to resist sandhi in stressed syllables, with varying success. My analysis extends Lin’s by exploring how stress interacts with Tone 3 sandhi within the boundaries of disyllabic feet and morpho-syntactic domains.

1. Introduction

Mandarin is the most widely spoken variety of Chinese and is also known as Standard Chinese. The goal of this paper is to document and analyze how Tone 3 sandhi interacts with stress in Mandarin, using data recorded from a native speaker.

The consultant for this paper is Rongying Zhang, a native Mandarin speaker born and raised in Shenzhen, a southern coastal city in China. She moved to the United States at age 18 to study biology at UCLA. Rongying primarily speaks standard Mandarin in daily communication, occasionally using a Northeastern accent at home due to her parents' regional background. She grew up bilingual in Mandarin and English, continuing to use Mandarin with family and friends while using English predominantly at UCLA.

The primary source for this paper is Lin's *The Sounds of Chinese* (2007), which offers a detailed description of Mandarin phonology, including tone sandhi and stress. The next section summarizes Lin's analysis of Tone 3 sandhi before introducing my own data on stress interaction.

2. Lin's (2007) Description and Analysis

According to Lin, Mandarin language has four basic tones that differentiate meanings of words: high level (T1), high rising (T2), low falling-rising (T3), and high falling (T4). There is also a special tone T0 known as the neutral tone that occurs only in non-initial positions of word or phrase and is treated as toneless in notation.

Here are some examples of tones in Mandarin from Lin, demonstrating that tones are crucial part to interpreting words meanings.

(1) Basic tones in Mandarin

Tone Number	Pitch Pattern	Pitch Value	Example
1	High Level	55	[ma] ₅₅ mā ‘mother’ 妈
2	High Rising	35	[ma] ₃₅ má ‘hemp’ 麻
3	Low Falling-Rising	214	[ma] ₂₄₁ mǎ ‘horse’ 马
4	High Falling	51	[ma] ₅₁ mà ‘to scold’ 骂

All four tones are all able to combine with one another. However, the pitch level and contour of tones can change according to contexts. Lin mentions T3 reduction in which T3 are simplified or shortened when they are followed by another tone (p. 196). He gives the following rule and example:

(2) T3 Reduction: 214 → 21 / __ another tone

Example: lǎoyīng ‘eagle’ [lau]214 [jəŋ]55 → [lau]21 [jəŋ]55

An important phonological phenomenon of Mandarin is tone sandhi, particularly the tone

3 sandhi which Lin describes as followed:

(3) Tone 3 Sandhi: T3 → T2 / __ T3 (i.e. 21 + 214 → 35 + 214) (Iterative)

3.12 Iterative Advanced
Seen also elicitation report
section, Lin didn’t mention
the rule as iterative.

This means that in a sequence of 2 T3s, the first T3 first experience T3 reduction and then changes to T2. Here is an example that illustrates the change:

- (4) /ɕjau/214 /kou/214 xiǎogǒu ‘puppy’
[ɕjau]21 [kou]214 T3 Reduction
[ɕjau]35 [kou]214 T3 Sandhi

Tone 3 sandhi is more complicated in phrases or sentences that consist of a string of tone 3 words. This means that tone 3 sandhi is not simply dependent on being followed by another T3,

below are examples that illustrate some of the possible different surface tone patterns that are produced by the same sequence of T3s, taken from Lin.

- (5) mǐ lǎo shǔ hǎo ‘Mickey Mouse is good’

T3 T3 T3 T3 → T3 T2 T2 T3

- (6) gǒu yǎo lǎo shǔ ‘The dog bit the mice’

T3 T3 T3 T3 → T2 T3 T2 T3

To understand how tone 3 sandhi is applied in complex words and phrases, Lin introduces the relevant prosodic and morpho-syntactic factors in Mandarin. The basic domain that tone 3 sandhi operates on is constituted by disyllabic foot. The below are examples of how syllables (σ) can be grouped into foot:

- (7) a. $\sigma \sigma \sigma \sigma \rightarrow (\sigma \sigma) (\sigma \sigma)$
 b. $\sigma \sigma \sigma \rightarrow \sigma (\sigma \sigma) \rightarrow (\sigma \sigma \sigma)$
 c. $\sigma \sigma \sigma \rightarrow (\sigma \sigma) \sigma \rightarrow (\sigma \sigma \sigma)$
 d. $(\sigma \sigma) (\sigma \sigma) \rightarrow (\sigma \sigma \sigma \sigma)$

Lin also provides examples of how such grouping is applied in real T3 words sequences:

- (8) a. jiu jiu jiu jiu (jiu jiu) (jiu jiu) ‘nine, nine, nine, nine’
 T3 T3 T3 T3 → (T2 T3) (T2 T3)
 b. jiu jiu jiu jiu jiu (jiu jiu) (jiu jiu) jiu ‘nine, nine, nine, nine, nine’
 T3 T3 T3 T3 T3 → (T2 T3) (T2 T3) T3
 → (T2 T3) (T2 T2 T3)

These examples help establish that disyllabic feet are built from left to right as the last syllable in (8b) is left alone in the first round of grouping before joining the preceding foot to

form a three-syllable domain. Tone 3 sandhi is applied after the domains are formed. However, the grouping process in (8) only helps explain (6) but not (5), which requires the initial disyllabic foot grouping of the second and the third syllables to produce the correct result. To account for this kind of grouping, Lin introduces morpho-syntactic domains and asserts that foot building must start with the smallest morpho-syntactic domains, which is *lǎoshǔ* ‘mouse’ in the case of (5). With this, Lin summarizes the rule for foot building for tone 3 sandhi: build the disyllabic foot from left to right and incorporate an unfooted free syllable to its adjacent foot. The process applies first within the smallest domain to the subsequent larger domains step by step, and applies to the smallest domain of phrase to the whole phrase from left to right. The example in (5) can now be explained by the rule as followed, where domains are represented with square brackets:

- (9) [[mǐ [lǎo shǔ]] hǎo] ‘[[Mickey [Mouse]] is good]’
 T3 T3 T3 T3
 (T3 T3) Word: disyllabic foot; T3 sandhi
 (T3 T2 T3) Word: incorporation
 (T3 T2 T2 T3) Phrase: incorporation; T3 sandhi

Lin also states that tone 3 sandhi applies within every domain but is optional when two adjacent T3s belong to different prosodic domains. Expressions with domains crowded toward the right side are also likely to have multiple patterns compared to the usual single pattern for expressions with domains crowded toward the left side. Alternative additional tone sandhi usually appears in fast or casual speech in which a T3 optionally change into T2 before a T2 that

alternated from T3. Below is an example of how an expression can alternate differently based on the forming of domains in fast speech

(10)	[xiǎo [mǔ [lǎo hǔ]]]	‘[small [female [tiger]]]’
	T3 T3 (T2 T3)	Word: disyllabic foot; T3 sandhi
	T3 (T3 T2 T3)	Word: incorporation; no T3 sandhi
	(T2 T3 T2 T3)	Word: incorporation; T3 sandhi

or Fast Speech Option 1

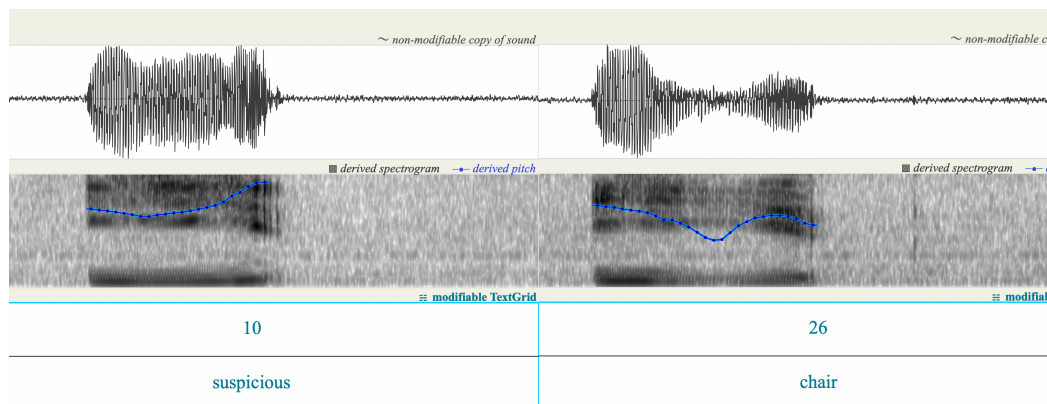
T3 (T3 T3 T3)	one prosodic domain for [mǔ [lǎo hǔ]]
T3 (T2 T2 T3)	T3 sandhi from left to right
(T3 T2 T2 T3)	Word: incorporation; no T3 sandhi

or Fast Speech Option 2

(T3 T3 T3 T3)	one prosodic domain for all syllables
(T2 T2 T2 T3)	T3 sandhi from left to right

3. Tone 3 Sandhi and Stress in Rongying’s Speech

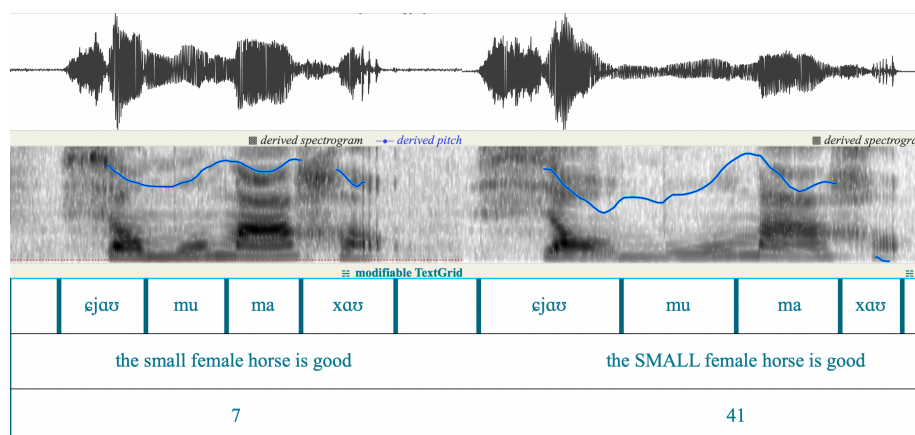
To better understand pitch difference between Tone 2 and Tone 3 in Rongying’s speech for setting a baseline for investigation, we recorded her saying words that only differ in tone. The below example from #10 [ji]35 ‘suspicious’ and #26 [ji]214 ‘chair’ demonstrate the difference between tone 2 and tone 3, with tone 2 showing a consistently rising pitch and tone 3 falling then rising.



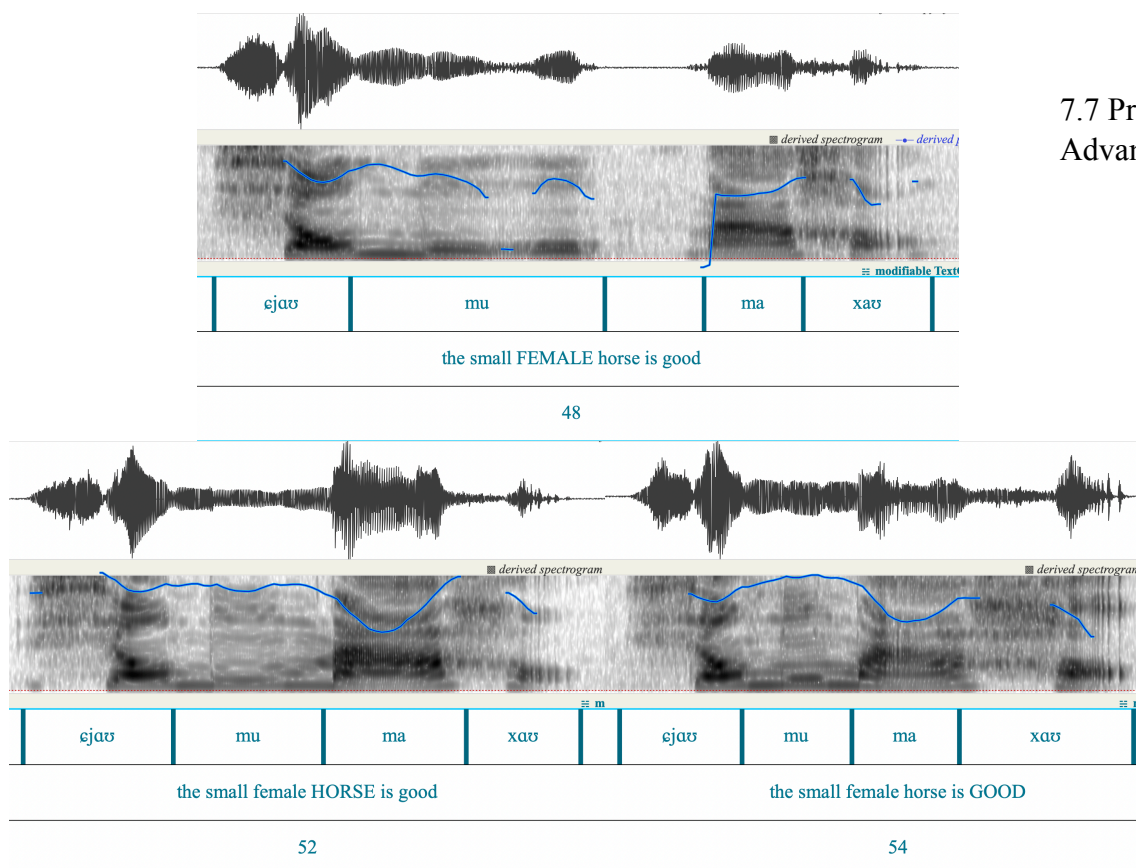
I also elicited words that demonstrate baseline T2 T3, Tone 3 Reduction, and T3 sandhi and their pronunciation aligned with expectations and description from Lin.

In order to investigate how stress influence Tone 3 sandhi, I ask Rongying to repeat phrases while stressing on different parts of the phrase. A notable example is the phrase #7 “the small female horse is good,” which provides a base case with all Tone 3 words. The most obvious comparison is between #7 and #52 where the word “horse” is stressed. As shown in #52, the pitch of “horse” demonstrate a clear T3 falling and rising pattern rather than the almost flat pattern in #7. #48 is the most unnatural one as my consultant had to take a pause after stress on “female” and showing a clear T3 pattern for it. #41 and #54 both show very similar pattern as the base case #7 which make sense because the proposed T3 sandhi form of #7 didn’t alternate the first word “small” and the last word “good” from T3 to T2.

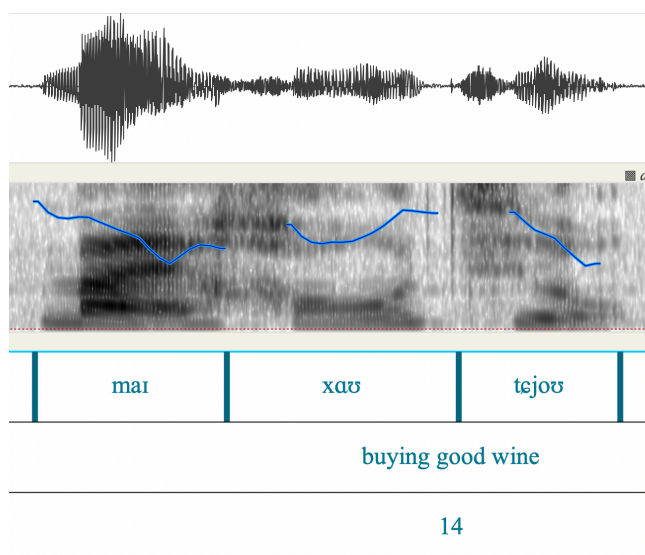
7.7 PrattScreenshot Advanced

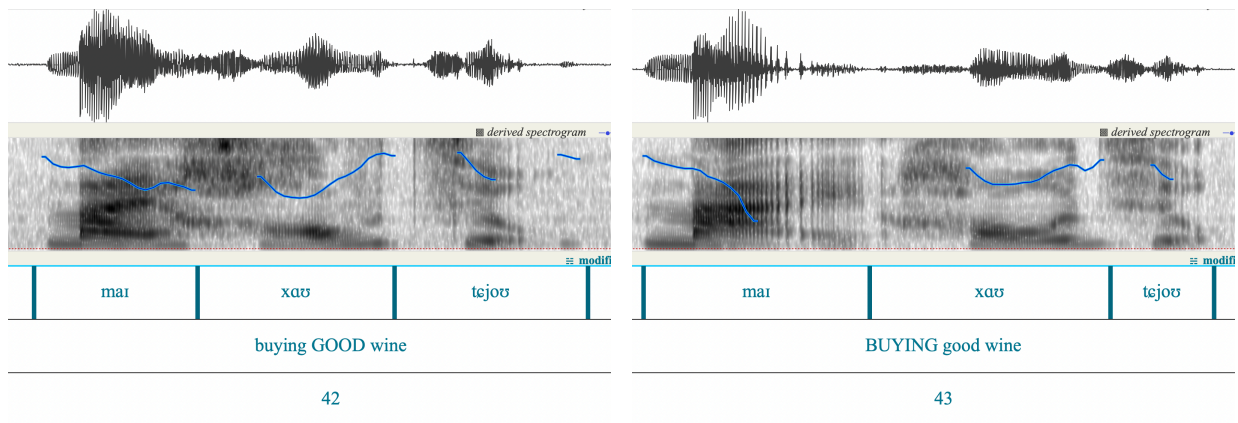


7.7 PrattScreenshot Advanced

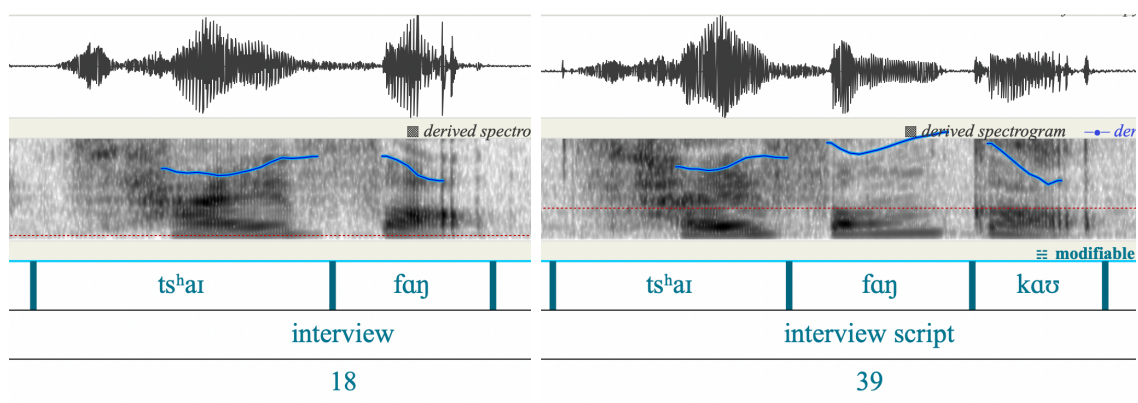


The change isn't as obvious in phrases like "buying good wine" from #14, #42, and #43. Despite stressing on 'GOOD' in #42, the speaker still pronounce the word with Tone 3 sandhi and show no clear difference with #14. However, one can still distinguish the change in pitch range with the [xaʊ] in #42 falling the lowest.





My theory is that the interaction between stress, whilst it can impact Tone 3 sandhi, are still restrained by a certain rule which makes trying not to apply Tone 3 Sandhi unnatural for speakers. As Lin observes, Tone 3 sandhi begins within the smallest morpho-syntactic domain and builds outward following a left-to-right foot structure. Rongying’s speech also verify this rule, shown by #18 [ts^hai35 faŋ214] ‘interview’ and #39 [ts^hai35 faŋ35 kau214] ‘interview script.’ While [faŋ214] is T3 in #18, it goes through an iterative Tone 3 Sandhi and alternates to T2 in #39 when it’s followed by another T3.



My data show that this morpho-syntactic domain constraint exerts strong influence even when speakers intentionally manipulate stress. In particular, stressed syllables within minimal morpho-syntactic domains (such as compounds or modifier-noun pairs) are more likely to

undergo sandhi regardless of emphasis. Thus, I conclude that the minimum morpho-syntactic domain serves as a primary trigger for Tone 3 sandhi, and that this constraint limits the speaker's ability to suppress sandhi in favor of stress-based distinctions. The smaller and more cohesive the domain, the more automatic and robust the application of sandhi, even when stress would suggest otherwise.

According to Lin, the smallest morpho-syntactic domain in the phrase from #7 is [mu214ma214] ([female horse]) which led to the following domain grouping:

(11)	[ɕjaʊ]214	[mu]214	[ma]214	[xaʊ]214	“the small female horse is good”
	T3	T3	T3	T3	underlying tone
		(T2	T3)		word: disyllabic foot; tone sandhi
	(T3	(T2	T3))		word: incorporation; no tone sandhi
	((T3	(T2	T2))	T3)	phrase: incorporation; tone sandhi

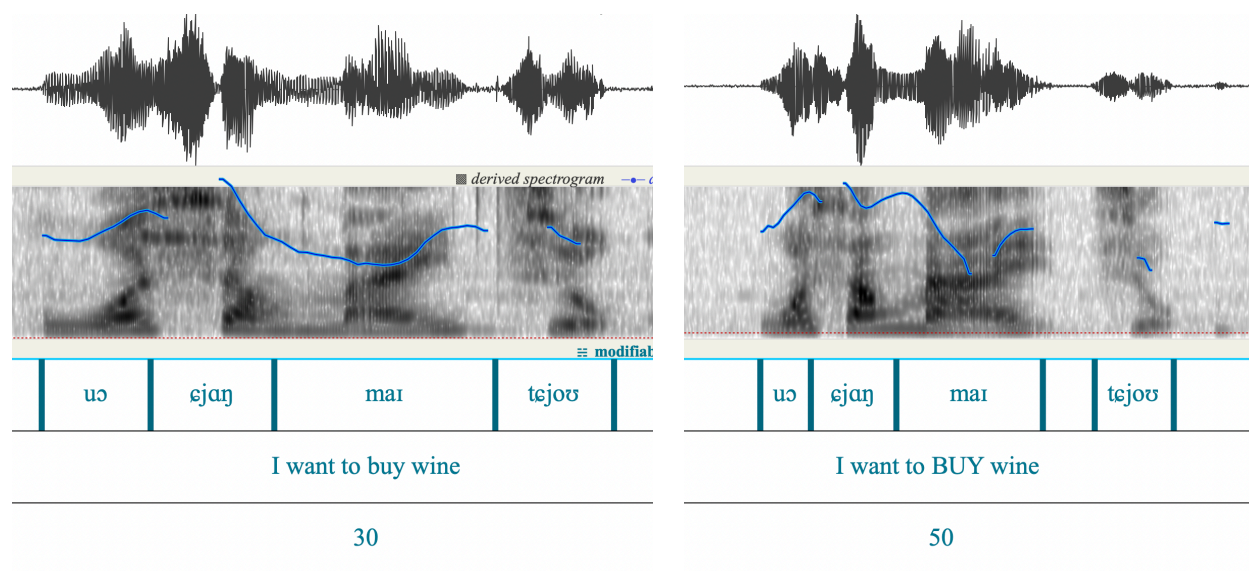
Because [mu214 ma214] “female horse” is the smallest morpho-syntactic domain where Tone 3 Sandhi applies first, it is more difficult for speakers to not apply tone sandhi. In Rongying's case, she has to stop between [mu214] and [ma214] when she attempts to stress [mu214] and pronounce it as a T3 word in #48. Similarly for #50 “I want to BUY wine.” The morpho-syntactic

brackets for the unstressed version (#30) is proposed by Lin is as followed:

(12)	[uə]214	[ɕjaŋ]214	[mai]214	[tɕjoʊ]214	“I want to buy wine”
	T3	T3	T3	T3	underlying tone
			(T2	T3)	phrase: disyllabic foot; tone sandhi
		(T3	(T2	T3))	phrase: incorporation; no tone sandhi
	(T2	(T3	(T2	T3)))	phrase: incorporation; tone sandhi

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Again in Rongying's speech, she had to stop between [mai]214 and [tɛjoʊ]214 in order to not apply tone 3 sandhi on [mai]214, the first word to undergo Tone 3 Sandhi in the smallest morpho-syntactic domain. This validates our theory.



4. Summary and Conclusion

My study of Tone 3 sandhi in Mandarin match Lin's findings. There are two new results. First, in Rongying's speech, Tone 3 sandhi applies naturally and consistently in neutral contexts but becomes less stable when the speaker attempts to impose stress. Second, stress-based resistance to sandhi is most likely to fail when the stressed syllable belongs to the smallest morpho-syntactic domain, suggesting that prosodic structure outweighs intentional stress. When speakers do successfully resist Tone 3 sandhi, they often insert a pause between words of the smallest morpho-syntactic domain, disrupting its structure .

A topic for future research isow the strength of morpho-syntactic boundaries affects speakers' ability to resist Tone 3 sandhi under focused stress. My findings suggest that sandhi is most robust when the target syllable is embedded within a tightly bound unit such as a compound

or modifier-noun pair, making resistance through stress difficult or unnatural. However, it remains unclear whether more loosely connected syntactic structures, such as across clause boundaries or in coordinate phrases, would show greater flexibility. A systematic comparison of sandhi behavior across different morpho-syntactic configurations, using controlled elicitation with stress manipulation, could clarify the gradient nature of domain effects and reveal whether certain syntactic groupings permit more speaker control over sandhi suppression.

References

Ethnologue. 2025. Ethnologue: Languages of the World, Twenty-Eighth edition. (Ed.)
 Lin, Yen-Hwei. 2007. The Sounds of Chinese. Cambridge, UK; New York:
 Cambridge University Press

Elicitation Word List

#	Transcription	Gloss	Translation
1	[ni]35 [xɑʊ]214	‘hello’	你好
2	[ɥy]35 [san]214	‘umbrella’	雨伞
3	[lɑʊ]35 [ʂu]214	‘mouse’	老鼠
4	[tʂan]35 [lan]35 [kwan]214	‘exhibition hall’	展览馆
5	[ma]35	‘hemp’	麻
6	[lɑʊ]21 [jən]55	‘eagle’	老鹰
7	[ɛjaʊ]214 [mu]35 [ma]35 [xɑʊ]214	‘the small female horse is good’	小母马好
8	[ɛjaʊ]35 [ma]35 [pʰɑʊ] 214	‘the small horse is running’	小马跑
9	[mi]214 [lɑʊ]35 [ʂu]35 [xɑʊ]214	‘Mickey Mouse is good’	米老鼠好
10	[ji]35	‘suspicious’	疑
11	[ʂoʊ]35 [lɿŋ]214	‘chief’	首领
12	[tɕʰi]35 [taʊ]214	‘to pray’	祈祷

13	[ɥy]21 [wən]35	‘language’	语文
14	[mai]214 [xau̯]35 [tɕjo̯]214	‘buying good wine’	买好酒
15	[na]35 [ʂʊŋ]214 [ko̯]35 [xau̯] 214	‘which kind of dog is better ’	哪种狗好
16	[tɕʰi]35 [ma]214	‘to ride a horse’	骑马
17	[li]35 [ɕjaŋ]214	‘ideal’	理想
18	[tsʰai]35 [faŋ]214	‘to interview’	采访
19	[ma]214	‘horse’	马
20	[uɔ̯]35 [ta]214 [la̯]35 [ʂu]214	‘I hit the mouse’	我打老鼠
21	[pa]35	‘pull out’	拔
22	[ljo̯]35 [ɕqœ]214	‘to bleed’	流血
23	[pa]214	‘handle’	把
24	[tʂau̯]214 [ʂo̯]35 [liŋ]214	‘finding the chief’	找首领
25	[ɥy]35 [fa]214	‘syntax’	语法
26	[ji]214	‘chair’	椅
27	[ta]214 [la̯]35 [xu]214	‘hitting the tiger’	打老虎
28	[tʂau̯]214 [tʂan]35 [lan]35 [kwan]214	‘looking for the exhibition hall’	找展览馆
29	[xwei]214	‘destry’	毁
30	[uɔ̯]35 [ɕjaŋ]214 [mai]35 [tɕjo̯]214	‘I want to buy wine’	我想买酒
31	[kʰɿ]35 [i]214	‘be capable of’	可以
32	[la̯]35 [pja̯]214 [xən]35 [xau̯]214	‘the old watch is very good’	老表很好
33	[ɕjo̯]35 [tʂaŋ]214	‘bear hand’	熊掌
34	[jʊŋ]35 [qœn]214	‘always’	永远
35	[ɕi]35 [tsau̯]214	‘to shower’	洗澡
36	[tʂʊŋ]35 [li]214	‘to tidy’	整理
37	[xwei]35	‘return’	回

38	[ɥy]21 [ji]51	‘semantics’	语义
39	[tsʰai]35 [faŋ]35 [kaʊ]214	‘interview script’	采访稿
40	[koʊ]214 [xən]35 [xaʊ]35 [jaŋ]214	‘dogs are easy to raise’	狗很好养
41	[ɕjaʊ]214 [mu]35 [ma]35 [xaʊ]214	‘the SMALL female horse is good’	小母马好
42	[mai]214 [xaʊ]35 [tɕjoʊ]214	‘buying GOOD wine’	买好酒
43	[mai]214 [xaʊ]35 [tɕjoʊ]214	‘BUYING good wine’	买好酒
44	[laʊ]35 [pjaʊ]214 [xən]35 [xaʊ]214	‘the OLD watch is very good’	老表很好
45	[uʊ]214 [ɕjaŋ]214 [mai]35 [tɕjoʊ]214	‘I want to buy wine’	我想买酒
46	[laʊ]35 [pjaʊ]214 [xən]214 [xaʊ]214	‘the old watch is VERY good’	老表很好
47	[uʊ]35 [ɕjaŋ]214 [mai]35 [tɕjoʊ]214	‘I WANT to buy wine’	我想买酒
48	[ɕjaʊ]214 [mu]214 [ma]35 [xaʊ]214	‘the small FEMALE horse is good’	小母马好
49	[mi]214 [laʊ]35 [ʂu]35 [xaʊ]214	‘Mickey Mouse is GOOD’	米老鼠好
50	[uʊ]35 [ɕjaŋ]214 [mai]214 [tɕjoʊ]214	‘I want to BUY wine’	我想买酒
51	[mi]214 [laʊ]35 [ʂu]214 [xaʊ]214	‘Mickey MOUSE is good’	米老鼠好
52	[ɕjaʊ]214 [mu]35 [ma]214 [xaʊ]214	‘the small female HORSE is good’	小母马好
53	[uʊ]35 [ɕjaŋ]214 [mai]35 [tɕjoʊ]214	‘I want to buy WINE’	我想买酒
54	[ɕjaʊ]214 [mu]35 [ma]35 [xaʊ]214	‘the small female horse is GOOD’	小母马好
55	[kʰɿ]35 [ni]214 [tʂaʊ]35 [uʊ]214 [mai]35 [pai]35 [wan]214 [mi]35 [fən]214	But you ask me to buy a hundred bowls of rice noodles.	可你找我买百碗米粉
56	[uʊ]35 [ɕjaŋ]214 [tɕjɛn]35 [tsou]214 [ɕjaʊ]35 [njaʊ]214 [wei]35 [y]214	I want to pick up the small bird’s tail feather.	我想捡走小鸟尾羽

57	[tɕʰɪŋ]35 [ni]35 [tɕɑʊ]35 [ljoʊ]214 [tsu]35 [tɕan]214 [maɪ]35 [pa]35 [ɕjaʊ]214 [y]35 [san]214	Please can you buy a small umbrella from team leader Liu?	请你找柳组长买把小雨 伞
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